

ADAPTIVE CIVIC ROUTING INTELLIGENCE SYSTEM

“Report once. Route correctly. Resolve with accountability.”

“Location determines responsibility. AI assists. GIS decides.”

Municipal jurisdiction is mathematically resolved via PostGIS boundary containment. Generative AI accelerates citizen complaint categorization without ever deciding administrative dispatch.

CORE CIVIC INTELLIGENCE SEPARATION

GEMINI ASSISTIVE AI

Parses unstructured text, predicts tags (AI_SUGGESTED), and removes intake friction.

POSTGIS DETERMINISM

ST_Covers evaluates GPS points against active MultiPolygons for 100% auditable dispatch.

SPATIAL ROUTING ENGINE

PostgreSQL + PostGIS

Deterministic ST_Covers Containment

ASSISTIVE INTELLIGENCE

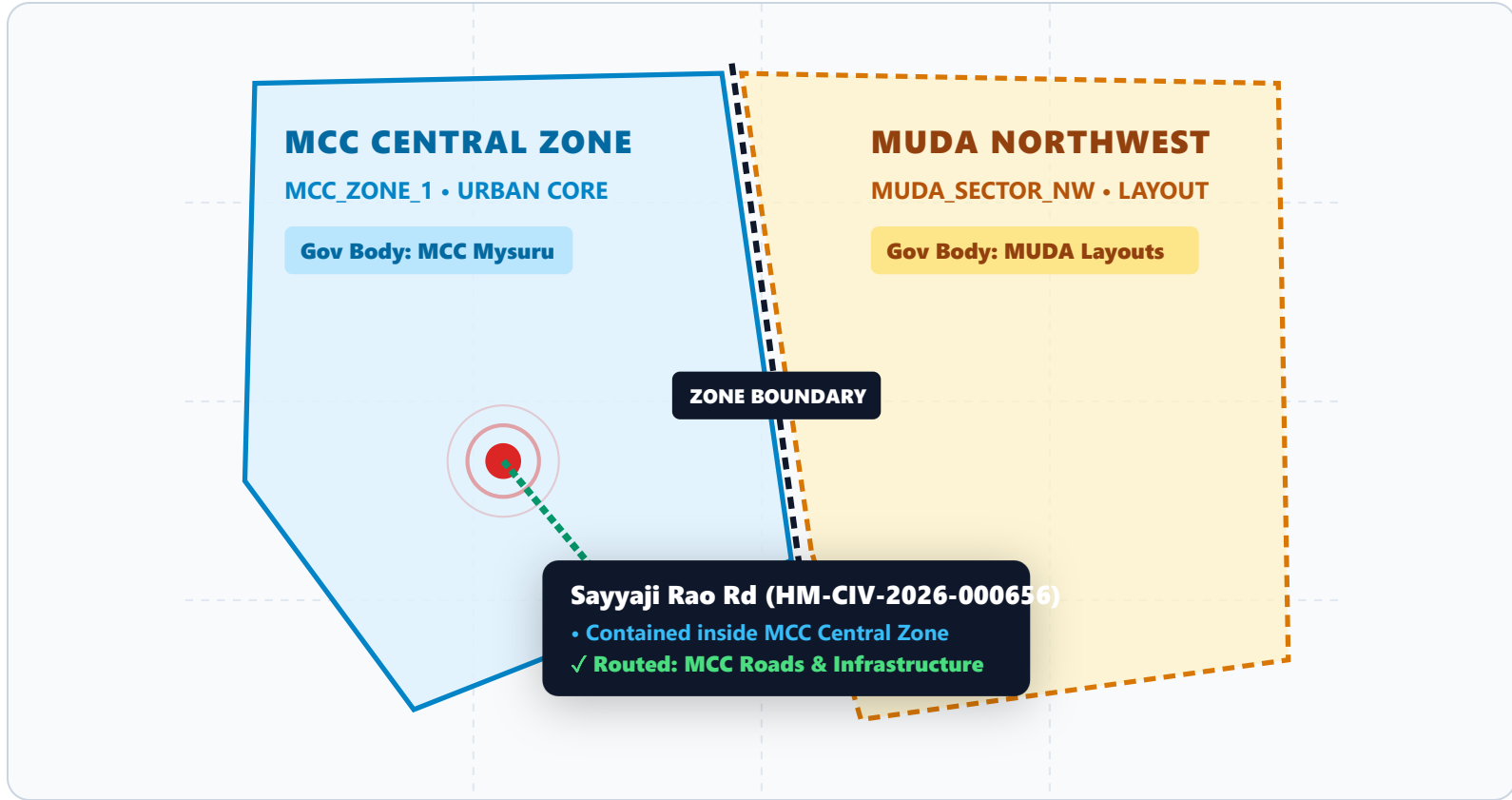
Google Gemini API

Category Suggestions (AI_SUGGESTED)

MYSURU REGION GIS CADASTRE

Automated Municipal Boundary Resolution

ACTIVE CADASTRE: MYS_2026_V2



Coordinate Reference: EPSG:4326 (WGS 84)

✓ Deterministic Spatial Point-in-Polygon Containment

1 Citizen Report
GPS Pin + Description + Photo

2 GIS Containment
PostGIS ST_Covers Match

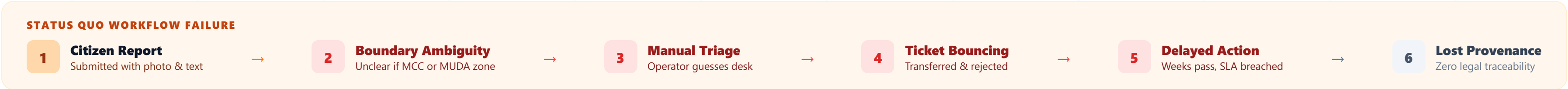
3 Correct Authority
MCC_ROADS Queue Bound

4 Field Action
Operator Claims & Dispatches

5 Audited Closure
Resolution Sealed in Ledger

Why Civic Complaints Get Stuck

“A citizen can report an issue correctly — yet the complaint can still reach the wrong authority.”



01

Boundary Ambiguity

Administrative boundaries are not obvious from a citizen's street knowledge or postal address.

Colloquial street names cross ward and layout borders. Pincodes encompass multiple municipal zones, creating an inherent geographic disconnect between citizens and civic administrators.

• Citizens describe landmarks; municipal systems operate strictly on cadastral MultiPolygons.

02

Manual Triage

Operators must determine which authority owns the location by guessing or cross-checking paper maps.

Intake desks are flooded with complaints requiring manual inspection. Clerks inevitably route tickets based on personal assumption, resulting in unpredictable queues.

• Creates intake bottlenecks, high labor overhead, and subjective dispatch decisions.

03

Routing Errors

A complaint can reach the wrong zone/department and require reassignment, resetting SLA timelines.

When a ticket arrives at the wrong office, it is rejected back to the triage pool. Each handover resets countdown clocks, leading to public frustration and zero resolution.

• Tickets bounce between MCC and MUDA desks with zero inter-agency accountability.

04

Lost Provenance

When boundaries change, historical routing decisions need to remain explainable and auditable.

Municipal wards are regularly reorganized or redrawn. If database records are updated in-place, historical tickets lose legal context and cannot be defended during public audits.

• In-place database table overwrites destroy past legal compliance records.



THE HACKATHON QUESTION

“How do we determine the correct authority from location — reliably and reproducibly?”

Deterministic Spatial Computing

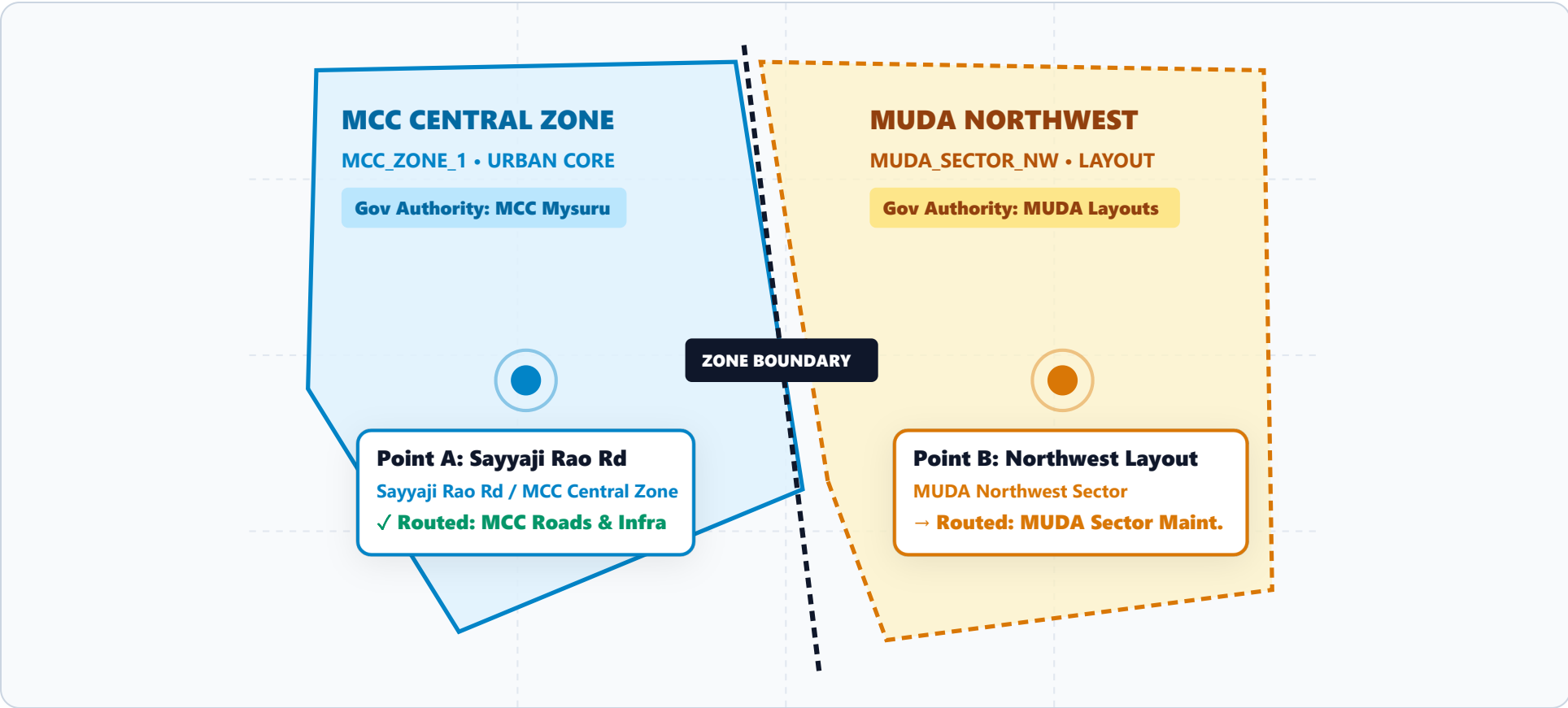
Deterministic Authority Resolution

Location Determines Responsibility

“The same civic issue can belong to different authorities depending on where it occurs.”

SPATIAL JURISDICTION DELINEATION

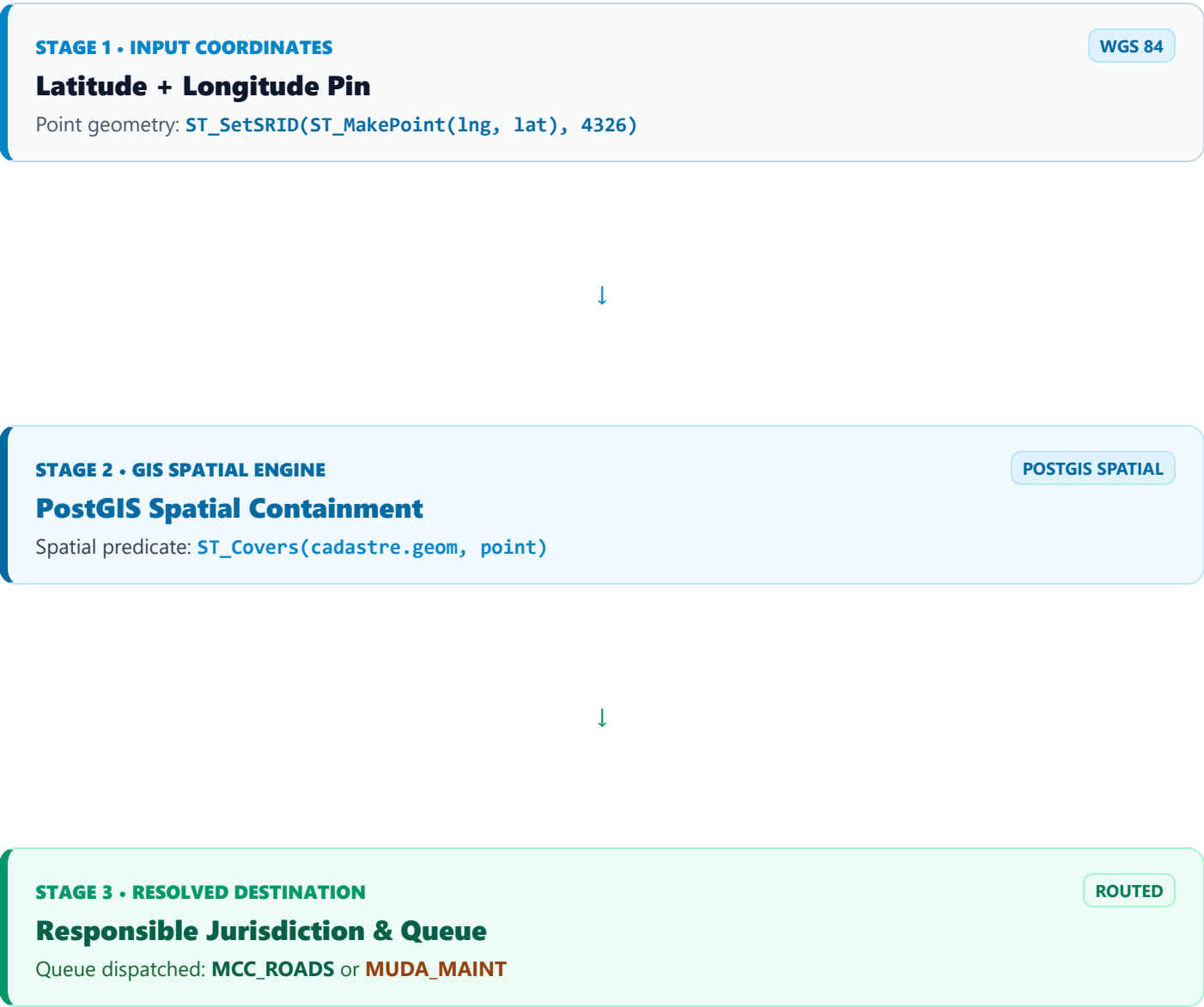
Two Identical Potholes • Two Completely Different Authorities



Zero keyword guessing

Spatial Containment (ST_Covers) Resolves True Authority

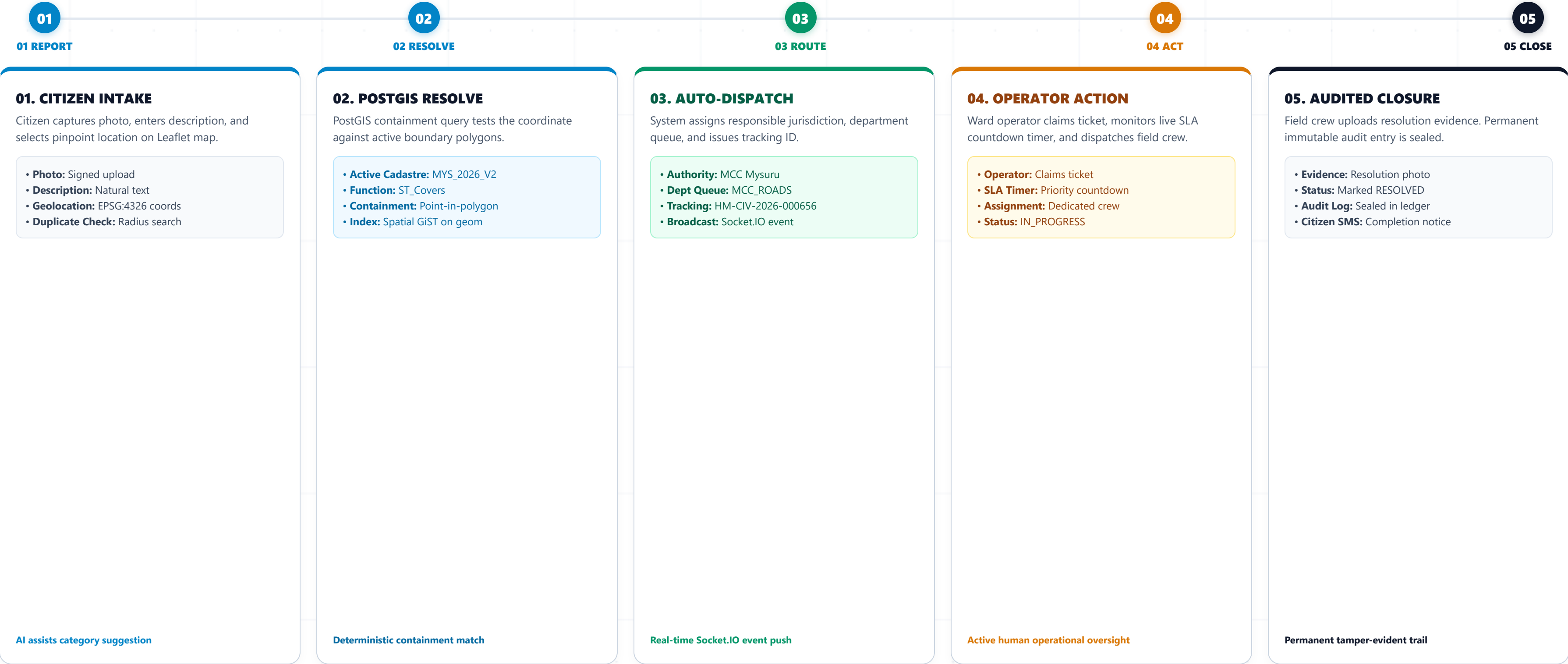
DETERMINISTIC ROUTING TRANSFORMATION



FUNDAMENTAL TAKEAWAY
“Text can describe a complaint. Coordinates determine where responsibility lies.”
This is fundamentally a spatial containment problem, not an AI chatbot problem. Coordinates mathematically dictate the responsible body.

From Citizen Report to Responsible Field Team

A seamless 5-stage pipeline delivering automated spatial routing, operator SLA tracking, and sealed audit closure.



AUTOMATED SYSTEM CORE

Complaint → Coordinate → Polygon → Jurisdiction → Department Queue

PostGIS spatial engine automatically matches coordinates against active cadastre polygons in sub-second queries.

ZERO HUMAN GUESSWORK

HUMAN CONTROL & OVERSIGHT

Operator Review → SLA Countdown → Field Action → Evidence Upload → Closure

Critical Guardrail: AI is NOT the routing authority. Operators retain triage control while GIS provides deterministic routing.

ACCOUNTABILITY

AI Assists. GIS Decides.

Clear architectural boundaries: Generative AI interprets citizen text; PostGIS deterministically decides civic jurisdiction.

ASSISTIVE INTELLIGENCE

GOOGLE GEMINI API

Complaint Classification

Natural language understanding to assist citizens in structuring complaints without bureaucratic friction.

- Smart Category Tagging:**
Suggests tags like "Pothole", "Garbage", or "Streetlight" labeled as **AI_SUGGESTED**.
- Structured Attribute Extraction:**
Extracts severity indicators and clean summaries from colloquial citizen text inputs.
- Intake Friction Reduction:**
Eliminates tedious manual navigation through complex municipal department hierarchies.

```
// Gemini Classification Payload
{ "category": "POTHOLE", "confidence": 0.96, "routing": "DEFERRED_TO_GIS" }
```

⚠️ STRICT ARCHITECTURAL GUARDRAIL

“Gemini is never permitted to decide jurisdiction or dispatch authority.”

LLMs cannot perform topological spatial proofs. AI hallucinations are completely isolated from municipal dispatch.

VS

STRICT
SEPARATION
OF ROLES

DETERMINISTIC ROUTING ENGINE

POSTGRESQL + POSTGIS

Spatial Containment & Routing

Mathematical point-in-polygon resolution providing verifiable authority mapping without hallucination risk.

- Cadastral Containment (ST_Covers):**
Tests coordinates mathematically against active MultiPolygon ward and zone boundaries.
- Deterministic Authority Mapping:**
Binds verified polygon results directly to official municipal department queues (**MCC_ROADS**).
- Auditable Execution Chain:**
Every routing decision is mathematically reproducible in a court of law or public audit.

```
-- Deterministic PostGIS Query
SELECT dept_code FROM zones WHERE ST_Covers(geom, ST_SetSRID(pt, 4326)); → 'MCC_ROADS'
```

🔒 MUNICIPAL GOVERNANCE GUARANTEE

“Coordinate → Polygon → Jurisdiction → Department”

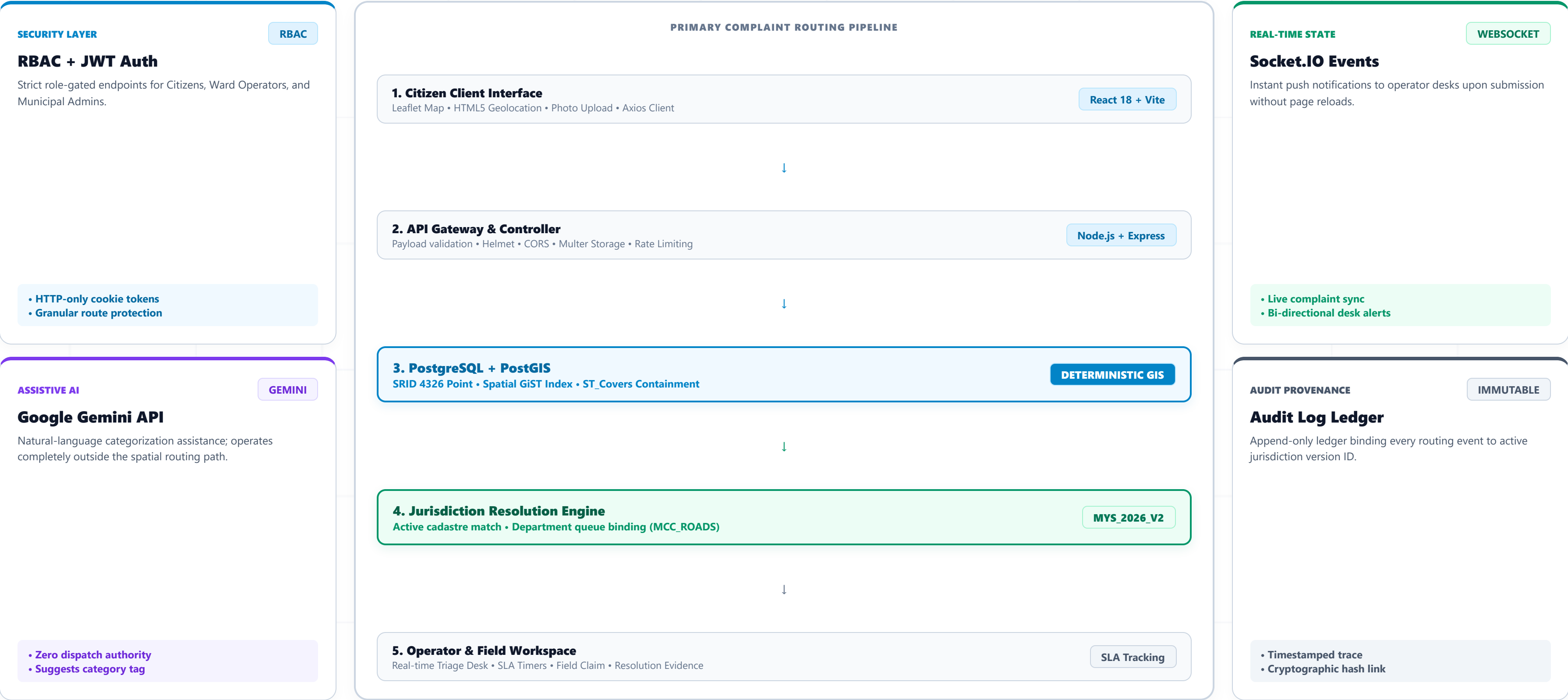
100% Deterministic logic. Zero risk of generative hallucination altering civic dispatch.

AI interprets. GIS determines responsibility.

“AI can assist interpretation without becoming the source of truth for civic jurisdiction.”

How the System Works Underneath

A modular full-stack GovTech architecture placing a deterministic spatial database at the center of civic routing.



POSTGIS TECHNICAL CORE:

• EPSG:4326 (WGS 84) Coordinate Reference

• Spatial GiST Indexing

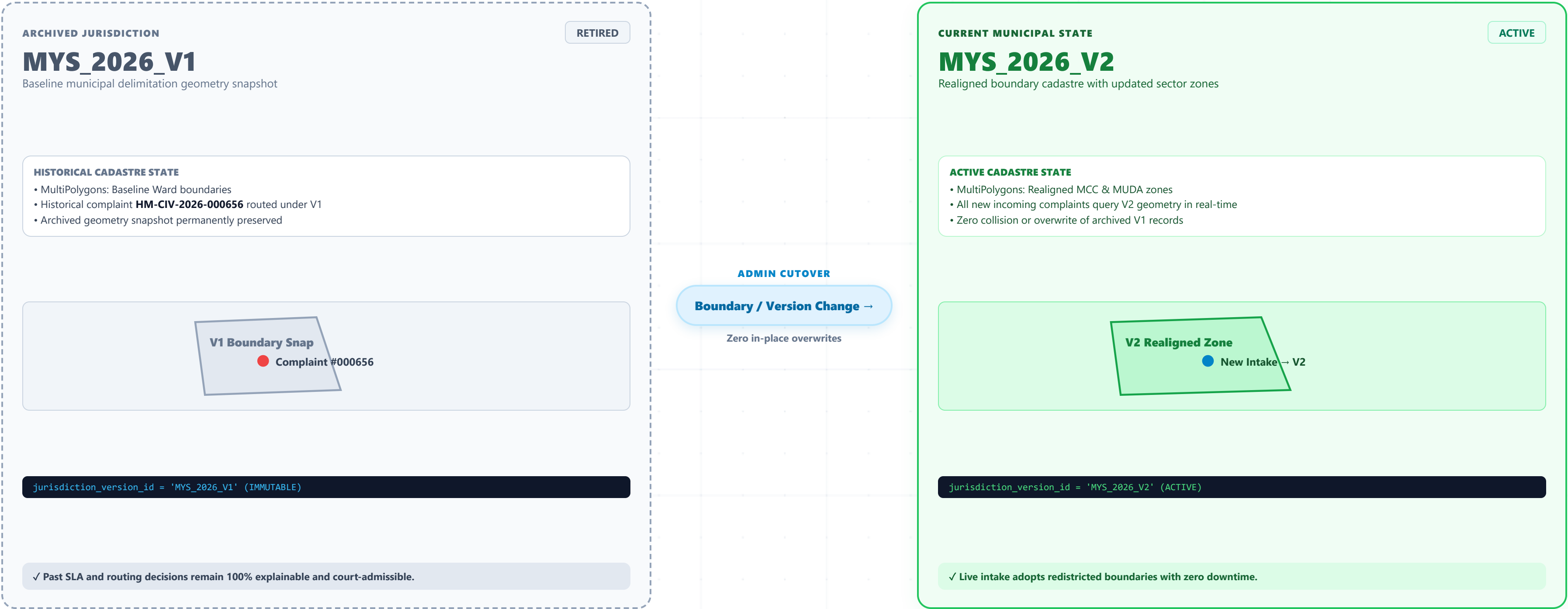
• ST_Covers MultiPolygon Containment

• Active Version Lookup

Deterministic Engine

Municipal Boundaries Change. History Cannot.

How immutable jurisdiction versioning preserves legal auditability when municipal wards are redrawn or renamed.



Built. Deployed. Verified.

Genuine application execution proof: Sayyaji Rao Road complaint routed via deterministic PostGIS logic.

GENUINE COMPLAINT PHOTOGRAPH

LIVE INTAKE EVIDENCE



CANONICAL COMPLAINT:
HM-CIV-2026-000656

VERIFIED LOCATION:
Sayyaji Rao Rd / MCC Central Zone

CATEGORY DETECTED:
POTHOLE (AI_SUGGESTED)

ACTIVE CADASTRE:
MYS_2026_V2 (ACTIVE)

✓ Responsible Department: MCC Roads & Infrastructure (MCC_ROADS)

VERIFIED SYSTEM EXECUTION TRACE

AUDIT LEDGER SEALED

01 Citizen Submission Intake
REST API: POST /api/complaints • Payload validated with coordinates

HTTP 201

02 Spatial Point Construction
PostGIS: ST_SetSRID(ST_MakePoint(Ing, lat), 4326) • Sayyaji Rao Rd / MCC Central Zone

SRID 4326

03 Jurisdiction Containment Resolved
Bound to MCC Central Zone via ST_Covers(cadastre.geom, point)

MATCHED

04 Department Queue Assignment
Routed: Roads & Infrastructure (MCC_ROADS) • Tracking code assigned

ROUTED

05 Operator Workspace Sync
Pushed live via Socket.IO; SLA priority countdown initiated

SLA TRACKED

06 Immutable Audit Ledger Record
Logged with timestamp, version hash, and actor ID into immutable ledger

SEALED

LIVE DEPLOYED MVP:

<https://adaptive-civic-routing.vercel.app>

Designed for Real-World Constraints

Demonstrating architectural maturity: Transparent analysis of current limitations and future scaling roadmaps.

01 / SPATIAL COMPUTE GIS Compute at Scale GIS BOTTLENECK IMPLEMENTED MVP Spatial GiST R-Tree indexing on boundary polygons and point geometries with deterministic <code>ST_Covers</code> containment queries. Sub-second resolution across municipal sets. CURRENT LIMITATION Evaluating multi-vertex boundary MultiPolygons with complex multi-vertex geometries across city-wide zones incurs compute overhead during high-concurrency disaster bursts (e.g. flash flooding). FUTURE SCALING ROADMAP Pre-computed quadkey spatial tiling, <code>ST_Subdivide</code> polygon simplification for bounding box filtering, and in-memory Redis caching of active zone centroids.	02 / FIELD CONNECTIVITY Offline Field Operations NETWORK LIMITS IMPLEMENTED MVP Graceful server degradation, network disconnection status banners, and reconnect polling for field operators. Server-side idempotency keys preventing duplicate ticket submissions. CURRENT LIMITATION Field workers inspecting subterranean drainages or dense markets frequently encounter mobile data blackouts, blocking immediate dispatch updates and sync. FUTURE SCALING ROADMAP Service Worker Progressive Web App (PWA) with client-side IndexedDB queue, cryptographic local change signatures, and store-and-forward sync upon reconnect.	03 / DATA VOLUME Database Scaling Ceiling DATABASE SCALE IMPLEMENTED MVP Node.js <code>pg.Pool</code> connection management (max 20 clients), normalized PostgreSQL schema, and foreign keys enforcing versioned boundaries. CURRENT LIMITATION A single PostgreSQL instance encounters write saturation during city-wide monsoon crisis complaint surges when thousands of citizens log reports simultaneously. FUTURE SCALING ROADMAP PostgreSQL streaming replication with read replicas for municipal analytics, time-series audit log partitioning (<code>pg_partman</code>), and zone-based database shards.
---	---	--

“Transparent engineering limits are part of the architecture — not hidden assumptions.”

Architectural Integrity

From Ward-Level Pilot to City-Wide Civic Intelligence

Scaling Mysuru's municipal governance with spatial precision, multilingual accessibility, and transparent public trust.

STRATEGIC ROADMAP PHASES

01

PHASE 1: WARD-LEVEL LIVE PILOT (CURRENT MVP)

Current verified routing foundation deployed in MCC Central Zone with local sanitation & road engineering crews. Validates response SLAs and duplicate detection.

- Deployed in MCC Zone 1
- 100% deterministic PostGIS containment
- Real Sayyaji Rao Rd verified

02

PHASE 2: FULL MUNICIPAL GIS INTEGRATION

Expand jurisdiction datasets across all MCC zones, MUDA peripheral layouts, and inter-agency utility routing (CESC Electricity, CADA Irrigation, Vani Vilas Water Works).

- Cross-agency jurisdiction datasets
- Automated inter-departmental escalation matrix

03

PHASE 3: KANNADA VOICE + TEXT INTERACTION

Voice-to-text complaint logging in Kannada for grassroots accessibility via WhatsApp & IVR channels across rural-urban fringes, ensuring equitable civic access.

- Indic-Wav2Vec / Bhashini ASR pipeline
- Automated landmark-to-coordinate bounding box extraction

RESPONSIBLE AI DISCLOSURE & GOVERNANCE

Google Gemini API: Natural-language classification assistance & category suggestions only. Strictly zero dispatch authority.

Antigravity AI: Development acceleration, architecture verification, and testing tooling.

✓ **Immutable Governance Guarantee:** All municipal boundaries, spatial containment proofs, and department routing remain 100% deterministic PostGIS spatial queries.

GOVTECH PRODUCTION VERIFICATION

- Spatial Relational Cadastre (EPSG:4326)
- Role-Based Access Control (RBAC/JWT)
- Tamper-Evident Historical Versioning (MYS_2026_V2)

Live Deployed MVP: <https://adaptive-civic-routing.vercel.app>
GitHub Repository: github.com/kkaribasappakl-eng/adaptive-civic-routing
Team Spark: HM26-E9C5 • Hemanth N, Karibasappa KL, Pooja M, Soumya Doddamani